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X, y, test_size=0.25, random_state=23
)

# Train the Random Forest classifier
clf = RandomForestClassifier(random_state=23)
clf.fit(X_train, y_train)

# Make predictions
y_pred = clf.predict(X_test)

# Generate confusion matrix
cm = confusion_matrix(y_test, y_pred)

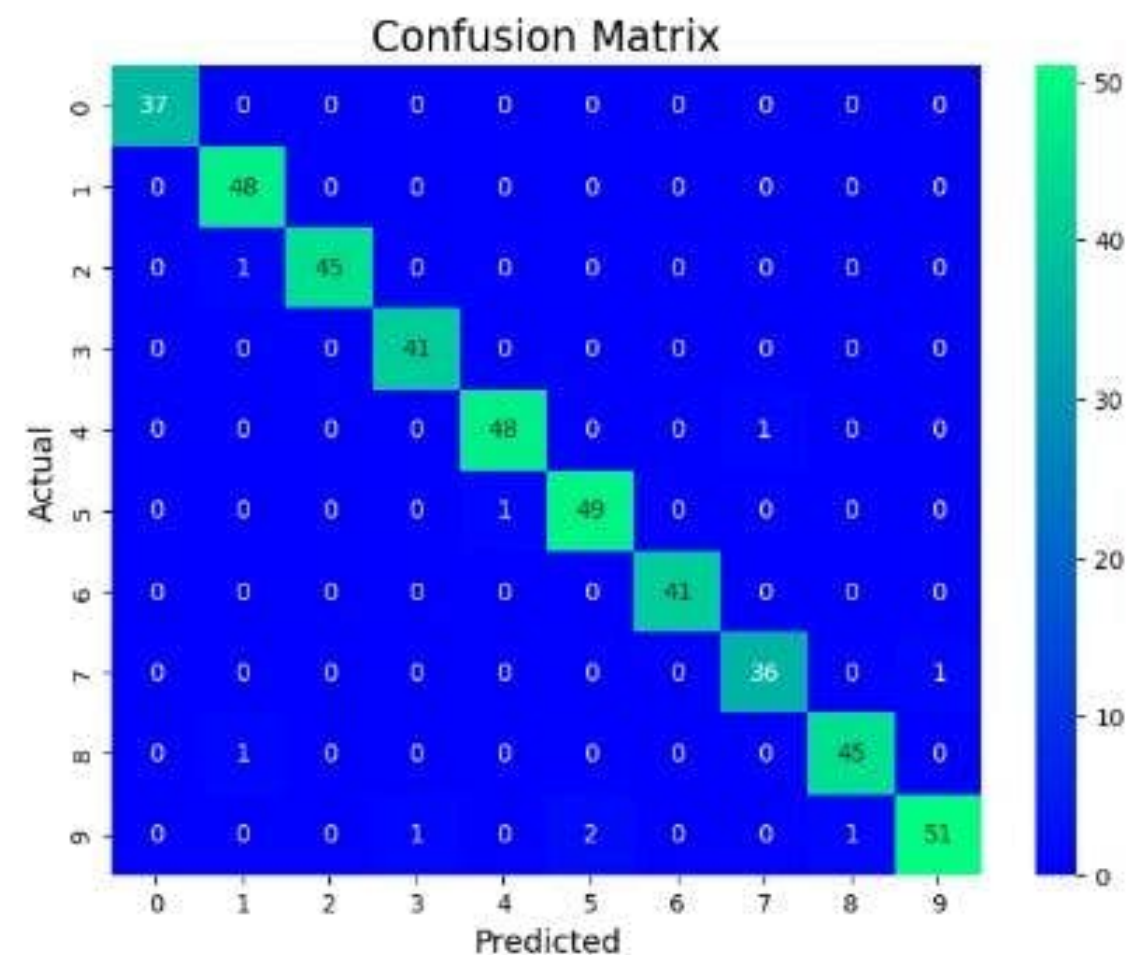
# Plot confusion matrix
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='winter')

plt.xlabel('Predicted', fontsize=13)
plt.ylabel('Actual', fontsize=13)
plt.title('Confusion Matrix', fontsize=17)
plt.show()

# Evaluation metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, average='weighted')
recall = recall_score(y_test, y_pred, average='weighted')
f1 = f1_score(y_test, y_pred, average='weighted')

# Print results
print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1 Score :", round(f1, 4))

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Accuracy : 0.98
Precision: 0.9802
Recall   : 0.98
F1 Score : 0.9799

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